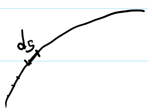
2025年4月14日 7:15

第17讲. 重积分几何应用 —曲面面积.

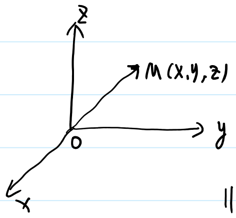
作业: P131. 1. 2①.②. 3. 4.



$$L:\begin{cases} x = x(t) \\ y = y(t) \\ z = z(t) \end{cases} \quad t \in [\alpha, \beta]. \quad \text{光滑.}$$

$$(x'(t),\ y'(t),\ z'(t)) \neq 0 .$$
$$\| \quad \vec{\tau}$$

微元法.



$$\|\overrightarrow{OM}\| = \sqrt{x^2+y^2+z^2}.$$

弧长微元 ds. (dl)

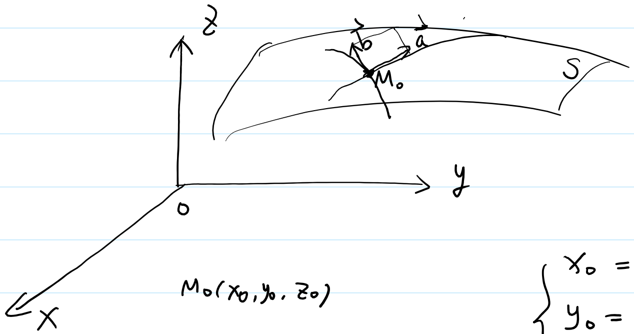
$$ds = \sqrt{(dx)^2+(dy)^2+(dz)^2} = \sqrt{(x'(t))^2+(y'(t))^2+(z'(t))^2}\, dt = \|\vec{\tau}\|\, dt$$

曲线段 L 弧长 $s = \int_\alpha^\beta \sqrt{x'(t)^2+y'(t)^2+z'(t)^2}\, dt$.

1. 空间曲面面积.

①. 曲面 S: $\begin{cases} x = x(u,v) \\ y = y(u,v) \\ z = z(u,v) \end{cases}$ $(u,v) \in D$. 光滑.

$x,\ y,\ z \in C^1(D)$. $\left.\dfrac{\partial(x,y,z)}{\partial(u,v)}\right|_{(u,v)}$ 秩为2.



$M_0(x_0,y_0,z_0)$

$(u_0,v_0) \in D$.

$$\begin{cases} x_0 = x(u_0,v_0) \\ y_0 = y(u_0,v_0) \\ z_0 = z(u_0,v_0) \end{cases}$$

$v = v_0$. $\begin{cases} x = x(u, v_0) \\ y = y(u, v_0) \\ z = z(u, v_0) \end{cases}$

$$\vec{r}(u,v) = \begin{pmatrix} x(u,v) \\ y(u,v) \\ z(u,v) \end{pmatrix}$$

$\vec{r}(u,v_0)$ 在 M_0 切向量

$$\vec{a} = \vec{r}_u'(u_0,v_0) = \left.\begin{pmatrix} x_u' \\ y_u' \\ z_u' \end{pmatrix}\right|_{(u_0,v_0)}$$

$u = u_0$. $\vec{r}(u_0,v) = \begin{pmatrix} x(u_0,v) \\ y(u_0, v) \end{pmatrix}$ $\vec{r}(u_0,v)$ 在 M_0 切向量

$\left.\begin{pmatrix} \\ z_u' \end{pmatrix}\right|_{(u_0,v_0)}$

$u=u_0$. $\vec{r}(u_0,v)=\begin{pmatrix} x(u_0,v) \\ y(u_0,v) \\ z(u_0,v)\end{pmatrix}$ $\vec{r}(u_0,v)$ 在 M_0 切向量

$$\vec{b}=\vec{r}_v'(u_0,v_0)=\left.\begin{pmatrix} x_v' \\ y_v' \\ z_v'\end{pmatrix}\right|_{(u_0,v_0)}$$

$\|\vec{a}\|du$

$$dS=\|\underline{\vec{a}du}\times\vec{b}dv\|=\|\vec{a}\times\vec{b}\|dudv$$

$\downarrow$

曲面面积微元.

$$=\|\vec{a}\|\cdot\|\vec{b}\|\cdot\sin\theta\, dudv$$

$$=\sqrt{\|\vec{a}\|^2\cdot\|\vec{b}\|^2\cdot(1-\cos^2\theta)}\ dudv$$

$$=\sqrt{\|\vec{a}\|^2\cdot\|\vec{b}\|^2-(\|\vec{a}\|\cdot\|\vec{b}\|\cos\theta)^2}\ dudv$$

$$=\sqrt{\|\vec{a}\|^2\cdot\|\vec{b}\|^2-\langle\vec{a},\vec{b}\rangle^2}\ dudv.=\sqrt{EF-G^2}\ dudv$$

$$\|\vec{a}\|^2=x_u'(u_0,v_0)^2+y_u'(u_0,v_0)^2+z_u'^2(u_0,v_0)=E$$

$$\|\vec{b}\|^2=x_v'(u_0,v_0)^2+y_v'(u_0,v_0)^2+z_v'(u_0,v_0)^2=F$$

$$\langle\vec{a},\vec{b}\rangle=x_u'(u_0,v_0)x_v'(u_0,v)+y_u'(u_0,v_0)y_v'(u_0,v_0)+z_u'(u_0,v_0)z_v'(u_0,v_0)=G.$$

$$\therefore\ \boxed{S=\iint_D\sqrt{EF-G^2}\ dudv}$$

$\downarrow$

曲面片面积.

② 曲面 S: $z=f(x,y)\in C^1(D)$.

$$S:\begin{cases} x=x \\ y=y \\ z=f(x,y)\end{cases}\quad (x,y)\in D.$$

$\vec{r}(x,y)=\begin{pmatrix} x \\ y \\ f(x,y)\end{pmatrix}$

$M_0(x_0,y_0,z_0)$

$z_0=f(x_0,y_0)$

$$\vec{a}=\vec{r}_x'(x_0,y_0)=\begin{pmatrix}1\\0\\f_x'(x_0,y_0)\end{pmatrix}\qquad \vec{b}=\vec{r}_y'(x_0,y_0)=\begin{pmatrix}0\\1\\f_y'(x_0,y_0)\end{pmatrix}.$$

$$\vec{a}\times\vec{b}=\begin{vmatrix} i & j & k \\ 1 & 0 & f_x'(x_0,y_0) \\ 0 & 1 & f_y'(x_0,y_0)\end{vmatrix}=k-f_x'(x_0,y_0)i-f_y'(x_0,y_0)j$$

$$=(-f_x'(x_0,y_0),\ -f_y'(x_0,y_0),\ 1)$$

$$\|\vec{a}\times\vec{b}\| = \sqrt{1+f_x'(x_0,y_0)^2+f_y'(x_0,y_0)^2}.$$

$$\therefore\ dS = \sqrt{1+f_x'(x_0,y_0)^2+f_y'(x_0,y_0)^2}\,dxdy$$

$$\therefore\ S = \iint_D \sqrt{1+f_x'^2(x,y)+f_y'(x,y)^2}\,dxdy$$

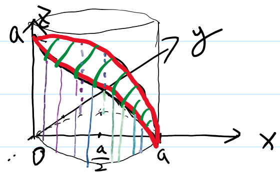
$$\vec{a}\times\vec{b} = \left.\begin{vmatrix} i & j & k \\ x_u' & y_u' & z_u' \\ x_v' & y_v' & z_v' \end{vmatrix}\right|_{(u_0,v_0)} = \Big(\underbrace{\left.\frac{D(y,z)}{D(u,v)}\right|_{(u_0,v_0)}}_{=A},\ \underbrace{\left.\frac{D(z,x)}{\partial(u,v)}\right|_{(u_0,v_0)}}_{=B},\ \underbrace{\left.\frac{D(x,y)}{D(u,v)}\right|_{(u_0,v_0)}}_{=C}\Big)$$

$$\|\vec{a}\times\vec{b}\| = \sqrt{A^2+B^2+C^2} = \sqrt{EF-G^2}.$$

例1. 计算 $x^2+y^2+z^2=a^2$ 被柱面 $x^2+y^2=ax$ 所截下的曲面的面积. $a>0$.

解:

①



$$\left(x-\frac{a}{2}\right)^2+y^2=\frac{a^2}{4}.$$

$$r^2 = ar\cos\theta.$$

$$z=\sqrt{a^2-x^2-y^2} \qquad D=\{(x,y)\mid x^2+y^2\le ax,\ y\ge 0\}.$$

$$S = 4\iint_D \sqrt{1+z_x'^2+z_y'^2}\,dxdy \qquad z_x' = \frac{-x}{\sqrt{a^2-x^2-y^2}}$$

$$= 4a\iint_D \frac{1}{\sqrt{a^2-x^2-y^2}}\,dxdy \qquad z_y' = \frac{-y}{\sqrt{a^2-x^2-y^2}}.$$

$$\sqrt{\frac{a^2}{a^2-x^2-y^2}}$$

$$\xlongequal[y=r\sin\theta]{x=r\cos\theta} \int_0^{\frac{\pi}{2}} d\theta \int_0^{a\cos\theta} \frac{r}{\sqrt{a^2-r^2}}\,dr$$

$$D_{r\theta} = \{(r,\theta)\mid \theta\in[0,\tfrac{\pi}{2}],\ r\in[0,a\cos\theta]\}$$

$$= 2a^2(\pi-2).$$

②

$$\begin{cases} x = a\cos\theta\sin\varphi \\ y = a\sin\theta\sin\varphi \\ z = a\cos\varphi \end{cases} \qquad \varphi\in[0,\tfrac{\pi}{2}], \quad \theta\in[0,\tfrac{\pi}{2}].$$

$$a^2\sin^2\varphi = a^2\cos\theta\sin\varphi. \quad \Rightarrow \sin\varphi = \cos\theta = \sin(\tfrac{\pi}{2}-\theta)$$

$$\varphi+\theta=\frac{\pi}{2}$$

$$\varphi + \theta = \frac{\pi}{2}$$

$$D_{\theta\varphi} = \{ (\theta, \varphi) \mid \theta \in [0, \tfrac{\pi}{2}],\ \varphi \in [0, \tfrac{\pi}{2}],\ \varphi + \theta \le \tfrac{\pi}{2} \}.$$

$$S = 4 \iint_{D_{\varphi\theta}} \sqrt{EF - G^2}\, d\varphi d\theta = 4 \iint_{D_{\varphi\theta}} a^2 \sin\varphi \, d\varphi d\theta = 4 \int_0^{\frac{\pi}{2}} d\theta \int_0^{\frac{\pi}{2}-\theta} a^2 \sin\varphi \, d\varphi = 2a^2(\pi - 2).$$

$$E = x_\varphi'^2 + y_\varphi'^2 + z_\varphi'^2 = a^2\cos^2\theta\cos^2\varphi + a^2\sin^2\theta\cos^2\varphi + a^2\sin^2\varphi = a^2$$

$$F = x_\theta'^2 + y_\theta'^2 + z_\theta'^2 = a^2\sin^2\theta\sin^2\varphi + a^2\cos^2\theta\sin^2\varphi = a^2\sin^2\varphi$$

$$G = x_\varphi' x_\theta' + y_\varphi' y_\theta' + z_\varphi' z_\theta' = 0$$

$\forall (x,y,z) \in \Omega$.　　$\rho(x,y,z)$.

dv　　$\forall (x,y,z)$.

$$dm = \rho(x,y,z)\, dv \qquad dV = dx\,dy\,dz.$$

$$M = \iiint_\Omega \rho(x,y,z)\, dv$$

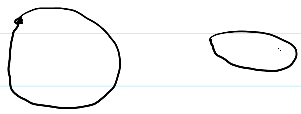
二. 曲线积分.

1. 可求长曲线.

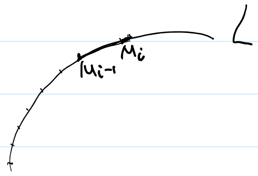
定义1. 设 L 是曲线. 一端 A. 另一端 B. 若规定 A 为起点. B 为终点

则 L 是一条有向曲线. 记为 $L(A,B)$ 或 $\overset{\frown}{AB}$

若起点和终点重合. 则称为封闭曲线.



没有自交点曲线 简单曲线.



$\overset{\frown}{M_{i-1}M_i}$　　$\overline{M_{i-1}M_i}$

$$\sum_{i=1}^{n} \Delta s_i \approx \sum_{i=1}^{n} \| \overline{M_{i-1}M_i} \|$$

定义2. 设 $L\subset\mathbb{R}^3$ 一条简单曲线段. $\begin{cases} x=x(t) \\ y=y(t) \\ z=z(t) \end{cases}$ $t\in[\alpha,\beta]$.

$\forall T=\{t_1, t_2, \cdots, t_{n-1}\}$. $t_0=\alpha$, $t_n=\beta$.

$$|T| = \max\{\Delta t_i = t_i - t_{i-1} \mid i=1,2\ldots n\}.$$

若 $\lim\limits_{|T|\to 0}\sum\limits_{i=1}^{n}\sqrt{(x(t_i)-x(t_{i-1}))^2+(y(t_i)-y(t_{i-1}))^2+(z(t_i)-z(t_{i-1}))^2}$ 存在.

且与划分方式 T 无关. 则 L 是可求长的.

$$ds = \sqrt{x'(t)^2+y'(t)^2+z'(t)^2}\,dt$$

$$s = \int_\alpha^\beta \sqrt{x'^2+y'^2+z'^2}\,dt$$